Generating “Truly Random” Digits Using Quantum Optics with Laser Manipulation

Madaly Gregory

Edmonds College

PHYS&222: Engineering Physics II with Lab

Dr. Rachel Wade

November 29, 2023

Abstract

Random numbers are essential to encryption, simulation, and randomized testing, but most computers generate only pseudorandom numbers, which are produced by an algorithm and can be reproduced from a known starting value. In the context of theoretical physics and its practical applications in cryptography and artificial intelligence, the distinction between true versus pseudorandomness is crucial in ensuring “random” sources are fundamentally unpredictable and free of bias. This experiment attempted to build a random digit generator based on a physical process. A red laser was split into two beams by a beamsplitter cube, each beam was directed onto a photoresistor, and an Arduino Uno compared the two voltages to produce a 0 or a 1. The generator did not produce random output. Across four trials (540 readings), each trial returned a single repeated digit, giving 35.4% zeros overall ($\chi^2 = 46.2$, $p < 10^{-9}$). A photon-flux estimate shows that the quantum noise at this laser power is about 3,000 times smaller than one step of the Arduino's analog-to-digital converter (ADC), so classical alignment differences determined every comparison. The photoresistors' slow response also made each reading depend on the previous one. This report quantifies these limitations and proposes a redesign using silicon photodiodes.

Introduction

The term "randomness" is associated with two significantly different concepts: phenomena that exhibit statistical randomness but are based on deterministic physical influence, and those that genuinely represent the *inherent* unpredictability associated with randomness. This distinction between true randomness and pseudorandomness is important in shaping the reliability and security of diverse applications in theoretical science and information security.

My research aims to examine this distinction by generating a source of truly random digits for practical use and verifying their inherent unpredictability through a statistical approach. The acknowledgment that true randomness is a valuable resource motivates an exploration of quantum behavior as a potential generator of sequences that transcend deterministic constraints.

Quantum mechanics is most widely accepted in the scientific community as a source of fundamental randomness. One proposed source of true randomness is in vacuum fluctuations–a unique quantum phenomenon with which no statistical predictability has been observed to date (Shi et al., 2016). These fluctuations describe the temporary changes in energy that occur at the subatomic level, even in a vacuum, which is typically thought of as devoid of matter and energy. According to the principles of quantum mechanics, empty space is not as empty as the vacuum classical physics defines. Instead, it is a field of virtual particles and antiparticles spontaneously popping in and out of existence (Reynaud et al., 2001). Fluctuations arise from Heisenberg's uncertainty principle–another principle of quantum mechanics–which states that there is a limit to the precision of which certain pairs of properties, such as position and momentum, can be simultaneously known. In the vacuum, particles and antiparticles can spontaneously emerge and borrow energy from the vacuum for extremely short periods, according to the uncertainty principle, before destroying each other. This activity sometimes results in spontaneous emission

of photons (Reynaud et al., 2001). For the context of this experiment, this behavior results in a random number and polarization of photons to be sourced. After decades of observation and statistical analysis, no evidence of physical constraints or patterns have been discovered to influence the rate of fluctuations in a subatomic vacuum, supporting the suggestion that this behavior is fundamentally non-deterministic. By relying on these fluctuations as a source of randomness and statistically analyzing the resulting digits to validate a lack of apparent patterns, the goal of this experiment is to validate the assumption of randomness as thoroughly as possible with available means. The goal was a simpler version built from parts available in a student lab: a laser, a beamsplitter, two light sensors, and an Arduino. The generated digits would then be evaluated statistically. The experiment did not succeed, but the cause can be identified quantitatively, which informs the design of a second iteration.

Materials and Methods

This experiment consisted of two main components: a physical circuit connected to an Arduino Uno, and a program to process the input and output the results in binary form. The circuit was built on a miniature breadboard. It relied on the voltage readings of two photoresistors connected in parallel. Both photoresistors received light input provided by a red laser diode powered by 5 volts, directed equally to both components through a 90-degree beamsplitter cube. Photons that are polarized parallel to the plane of incidence (horizontally at the center of the beamsplitter in this case) pass directly through. Those that are polarized orthogonal to this plane (and thus vertically polarized in this context) are deflected 90 degrees to the right. The photoresistors were placed in front of each resulting beam. Component holders were custom designed and 3D-printed to stabilize both the beamsplitter and laser in place along a level, horizontal axis. A 220-ohm resistor was placed in series with each photoresistor to allow for voltage measurement. During each iteration of the program, the laser diode was powered via a digital pin for a short interval. At the moment the power to the laser was cut, the voltages of each photoresistor were compared. Each of these readings was assigned a binary value of “0” or “1”, and the highest voltage reading of the two would be printed alongside the binary value corresponding to the photoresistor that provided the higher reading. If both were equal, then the function reiterated.

Hardware

Controller: Arduino Uno with a 10-bit analog-to-digital converter (ADC). With a 5 V reference, one count is about 4.88 mV.

Light source: 5 V red laser diode module (assumed ≤5 mW at roughly 650 nm), powered from a digital output pin.

Optics: A beamsplitter cube that separates light by polarization. Light polarized parallel to the plane of incidence (horizontal) is transmitted, and light polarized perpendicular to it (vertical) is deflected 90 degrees. Custom 3D-printed mounts held the laser and cube level and aligned.
Detectors: Two GL5528-type photoresistors, also called light-dependent resistors (LDRs). LDR 1 received the transmitted (horizontal) beam and LDR 2 the deflected (vertical) beam. Each LDR was placed in series with a 220 Ω resistor to form a voltage divider on 5 V, and the ADC measured the voltage across the resistor. The circuit was built on a miniature breadboard.
Environment: A dark room with a small amount of ambient light entering around the doorframe.

Firmware

In each loop iteration, the laser was pulsed on for a short interval (approximately 50 ms), and both LDR voltages were read in sequence with `analogRead` at the moment the laser was switched off. One `analogRead` takes about 112 µs on an Uno, so the second channel was sampled about 0.22 ms after the first. The program printed the higher voltage together with a bit: 0 if LDR 1 was higher and 1 if LDR 2 was higher. Equal readings triggered a repeat. The loop ran at roughly 10 readings per second, and the output was recorded over the serial port.

Results

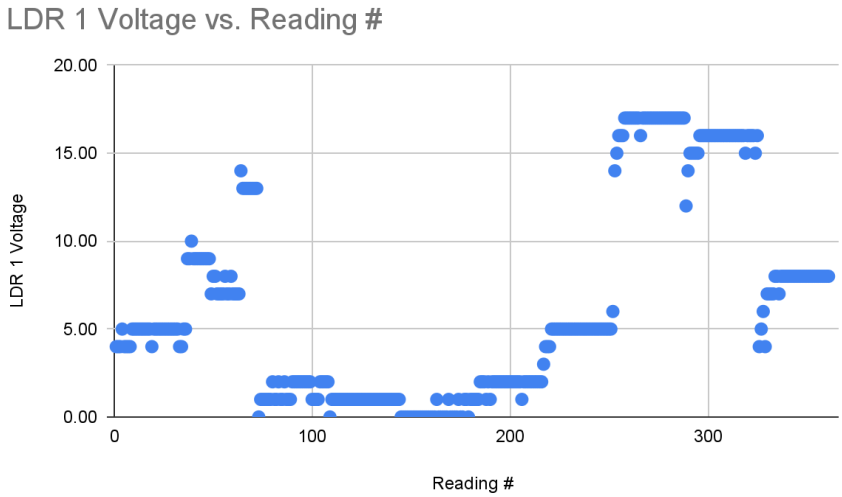


Figure 1. Voltage of LDR 1 (transmitted, horizontally polarized beam) versus reading number.

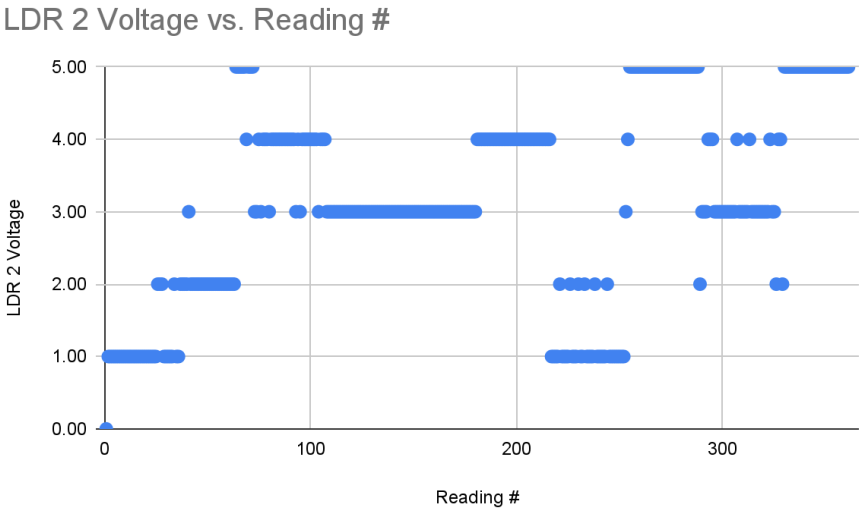


Figure 2. Voltage of LDR 2 (deflected, vertically polarized beam) versus reading number.

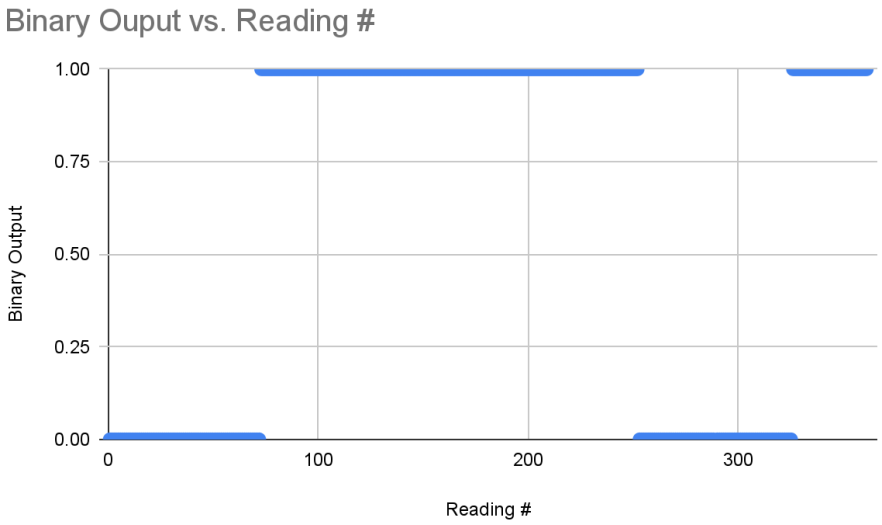


Figure 3. Binary output versus reading number. 0 = LDR 1 higher; 1 = LDR 2 higher.

Trial	Readings	N	Output	LDR 1 (V)	LDR 2 (V)	ΔV = V1 − V2 (V)
1	1–73	73	all 0	1.14	0.71	+0.43
2	74–253	180	all 1	0.69	1.05	−0.36
3	254–371	118	all 0	1.08	0.83	+0.25
4	372–540	169	all 1	0.77	1.01	−0.24

Table 1. Per-trial summary (voltages are means across the trial).

In every trial, one LDR read higher than the other on every reading, and the winning channel changed between trials (*Table 1*). In the plot of Binary Output vs. Reading (*Figure 3)*, each streak of the same value of either 0 or 1 represents a distinct trial time period in which readings were measured. In the first trial of readings (from reading numbers 1-73), the binary output was

consistently 0, indicating that LDR 1 (the LDR measuring horizontally-polarized photons) was receiving a higher number of photons, and thus reading a higher voltage. In the second trial of readings (from reading numbers 74-253), the binary output was consistently 1, indicating that LDR 2 (measuring vertically-polarized photons) was receiving a higher number of photons and thus reading a higher voltage. A similar effect is seen in the subsequent two trial periods, consistently providing values of solely 0 or 1.Within a trial, each channel's standard deviation was about 7 mV (roughly 1.4 ADC counts). The smallest difference between channels (0.24 V, about 49 counts) is roughly 34 times the noise within the trial, so the outcome was never close. These voltages correspond to LDR resistances of about 0.8 to 1.4 kΩ. For a GL5528, which is 10 to 20 kΩ at 10 lux with $\gamma \approx 0.7$, this implies several hundred lux at the sensor, which is consistent with a focused laser spot.

Overall, 191 of 540 bits were 0 (35.4%), compared with an expected 270 for a good source. The deviation of 79 counts is about 6.8 standard deviations ($\chi^2 = 46.2$, 1 degree of freedom, $p < 10^{-9}$). The bit stream is strongly biased, so it is not random. A wiring fault is an unlikely cause. This cannot be traced to a lack of power to either LDR or differing power supply, since both LDRs read above 0, were connected in parallel to the same 5V supply, and individually in series with one 220 Ω resistor each.

Discussion

This experiment has served as an entrypoint to using quantum optics as a means of harnessing true randomness for the creation of fundamentally non-deterministic data. After analyzing the results and experimental setup, several key areas of improvement and sources of error have been identified.

First, quantum noise was far below the measurement resolution. A 5 mW beam at 650 nm delivers about 1.6×10^{16} photons per second (about 3.06×10^{-19} J per photon). Each detector receives roughly half, or about 8×10^{12} photons in 1 ms. Shot noise increases with the square root of the photon count, so the relative fluctuation is $1/\sqrt{N} \approx 3.5 \times 10^{-7}$. One ADC step is about 1/1023, or 10^{-3}, of full scale, which makes the quantum fluctuation roughly 3,000 times smaller than the smallest step the Arduino can go down to. At this point, the comparison is decided by classical differences between the two channels. The vacuum fluctuations in Shi et al. (2016) are accessible only with a shot-noise-limited balanced homodyne detector, which this circuit is not.

The split ratio between the photoresistors depends on polarization angle. A polarizing cube divides a linearly polarized beam according to Malus's law, so the fraction of light that reaches each photoresistor depends on the rotation of the laser's polarization relative to the cube. At 45° the split is 50/50, but a 5° rotation shifts it to about 59/41, a difference of nearly 17% that is significantly larger than the noise. Small movements of the laser or cube between trials would change which channel received more light, which is consistent with the sign changes observed.

Most importantly, photoresistors have proven to be insufficient in measuring photon input. Photoresistors were originally chosen as they are easily accessible, cost-efficient, robust, and straightforward in their ability to provide voltage readings depending on the quantity of light input. After analyzing the results and further researching these components, it became evident that photoresistors display time-dependent characteristics and temperature sensitivity. These components can store photon measurements for a brief period after the input light source is removed (Louise, 2020). The GL5528 has a rise time of about 20 ms and a decay time of about 30 ms (Senba Sensing Tech, GL55 series table), a resistance that follows a power law with illuminance ($\gamma \approx 0.7$). The 20 ms response time is too slow for the sampling scheme. Voltages were read about 1 ms after the laser shut off. With a decay time constant near 30 ms, only about 3% of the signal has decayed by then ($e^{(-1/30)} \approx 0.967$), so each reading mostly reflects the sensor's stored state rather than the light at that instant.

Additionally, minor temperature fluctuations can affect voltage readings during measurement. Warmer temperatures lower resistance and thus force voltage readings to be higher than expected (Louise, 2020). Considering their unreliability, we suggest the photoresistors be replaced with a more sensitive light-measuring component that is free of temperature and time dependency. A silicon photodiode addresses most of these limitations, as they are significantly higher in sensitivity and can more accurately detect small phase shifts from the incoming laser beam (Rafaelli, 2018). The Thorlabs FDS100 is used here as an example because its specifications are published (Thorlabs, 2017). Table 2 compares it with the GL5528.

Lastly, joining resistors in series with the laser diode–reducing its power–should result in fewer photons reaching the detectors at once, thus preventing oversaturation and inaccurate readings.

Aside from the choice of components themselves, it is also worth considering the effect of ambient lighting on light readings in either case. It is difficult to calculate and account for the amount of photons that may have contacted the photoresistors from the openings around the doorframe. In future iterations of this experiment, securing the circuit inside a light-proof container and conducting measurements in a darker room will mitigate external light interference. A combination of sensitive photodiode replacements and a light proof setting should allow the circuit to detect vacuum fluctuations at smaller magnitudes with less classical interference.

Proposed redesign

1. Detectors: Use two photodiodes, each with a current-to-voltage amplifier, and attenuate the beam with a neutral-density filter to prevent saturation. Run the laser continuously instead of pulsing it.
2. Optics: Use a non-polarizing 50/50 beamsplitter, lock the laser's rotation in place, and enclose the setup in a lightproof box.
3. Firmware: Sample on a fixed schedule from a hardware timer (for example, 1 kHz) instead of a laser pulse delay loop.
4. Bit bias removal: Subtract the average difference between channels and take the sign. A von Neumann corrector (von Neumann, 1951), which examines pairs of readings, can remove residual bias, but it does not remove dependence between bits, so independence must still be tested.

5. Validation: Collect at least one million bits and evaluate them with the NIST statistical test suite (Bassham et al., 2010) and entropy estimation methods (Turan et al., 2018). Add a firmware check that flags a stuck or repeating output, since that was the failure observed here.
6. Detector sensitivity: A milliwatt laser is a classical light source. A defensible claim of quantum randomness would require a shot-noise-limited balanced homodyne detector (Shi et al., 2016) or a single-photon detector, and a measurement showing the electronic noise is sufficiently below the quantum noise.

Conclusion

The photoresistor and beamsplitter circuit did not produce truly random digits. Its output was influenced by classical differences between the two channels, with 35.4% zeros and 4 runs where about 248 were expected. A shot-noise estimate shows the quantum signal was about 3,000 times smaller than one ADC step, and the LDRs' 20 to 30 ms response added memory between readings. Replacing the LDRs with silicon photodiodes (10 ns response, 1 nA dark current), along with timer-based firmware and a scripted validation process, would result in a system worth testing. Demonstrating vacuum-fluctuation randomness would still require a shot-noise-limited homodyne detector.

Acknowledgements

Thank you to Edmonds College peer and materials science engineer Cayden Bock for helping me design the custom 3D-printed component holders, which held the laser and beamsplitter on a stable, level axis.

References

Louise. (2020, May 19). *Photoresistor: Basics and Arduino tutorial.* JAK Electronic. https://www.jakelectronics.com/blog/photoresistor-basics-and-arduino-tutorial

Raffaelli, F. (2018). *Quantum random number generators in integrated photonics* [Doctoral dissertation, University of Bristol]. https://research-information.bris.ac.uk/ws/portalfiles/portal/187187475/Final_Copy_2019_01_23_Raffaelli_F_PhD.pdf

Reynaud, S., Lambrecht, A., Genet, C., & Jaekel, M. T. (2001). Quantum vacuum fluctuations. *Comptes Rendus de l'Académie des Sciences – Series IV – Physics, 2*(9), 1287–1298. https://doi.org/10.1016/S1296-2147(01)01270-7

Senba Sensing Tech. (n.d.). *GL55 series CdS photoresistor specifications* (GL5528: 10–20 kΩ at 10 lux, 1 MΩ dark, γ = 0.7, 20 ms rise, 30 ms decay). https://piekarz.pl/pdf/4020

Shi, Y., Chng, B., & Kurtsiefer, C. (2016). Random numbers from vacuum fluctuations. *Applied Physics Letters, 109*(4). https://doi.org/10.1063/1.4959887

Thorlabs. (2017). *FDS100 Si photodiode specification sheet* (0637-S01, Rev. M). https://www.physics.utoronto.ca/~astummer/Archives/2011%20Balanced%20TIA/Docs/Thorlabs%20FDS100%20PD.pdf

Thorlabs. (n.d.). *Photodiode tutorial: Theory of operation* (dark current versus temperature). https://www.thorlabs.de/catalogpages/Obsolete/2020/FGAP71.pdf

Turan, M. S., Barker, E., Kelsey, J., McKay, K., Baish, M., & Boyle, M. (2018). *Recommendation for the entropy sources used for random bit generation* (NIST Special Publication 800-90B). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-90B